

Math F113X: Homework Set 2 — Solutions

Voting Theory (p. 53): # 12, 13–16, 18, 24ab, and Problem A
Weighted Voting (p. 70): # 2, 4, 5, 6

Part I: Voting Theory

Problem 12

The downtown business association is electing a new chairperson, and decides to use approval voting. The tally is below, where each column shows the number of voters with the particular approval vote. Which candidate wins under approval voting?

Number of voters	8	7	6	3	4	2	5
A	X	X			X		X
B	X		X	X			X
C		X	X	X		X	
D	X		X		X	X	

Solution

Under approval voting, each candidate's score is simply the total number of voters who marked an X in that candidate's row. Add across each row.

Candidate	# of approvals	
A	$8 + 7 + 4 + 5$	$= 24$
B	$8 + 6 + 3 + 5$	$= 22$
C	$7 + 6 + 3 + 2$	$= 18$
D	$8 + 6 + 4 + 2$	$= 20$

Candidate A has the most approvals, so the winner is Candidate A.

Problem 13–16

Problems 13–16 all refer to the same situation:

An election resulted in Candidate A winning, with Candidate B coming in a close second, and candidate C being a distant third.

(13) If for some reason the election had to be held again and C decided to drop out of the election, which caused B to become the winner, which is the primary fairness criterion violated in this election?

Solution

This violates the **Independence of Irrelevant Alternatives Criterion (IIA)**, because candidate C is irrelevant to the competition as a “distant” third. Removing a candidate who never had a chance of winning should not change which of the remaining candidates wins.

(14) If for some reason the election had to be held again and many people who had voted for C switched their preferences to favor A, which caused B to become the winner, which is the primary fairness criterion violated in this election?

Solution

This violates the **Monotonicity Criterion**. If A gains more support, that should not decrease A’s chance of winning — yet here A gained support and lost the election.

(15) If in a head-to-head comparison a majority of people prefer B to A or C, which is the primary fairness criterion violated in this election?

Solution

This violates the **Condorcet Criterion**, since B is preferred in a head-to-head comparison with all other candidates. A Condorcet candidate exists (namely B), but a different candidate (A) won.

(16) If B had received a majority of first place votes, which is the primary fairness criterion violated in this election?

Solution

This violates the **Majority Criterion** — a candidate receiving a majority of the first-place votes should be the winner, but B did not win.

Problem 18

In the election shown below under the Borda Count method, explain why voters in the second

column might be inclined to vote insincerely. How could it affect the outcome of the election?

Number of voters	20	18
1st choice	A	B
2nd choice	B	A
3rd choice	C	C

Solution

Recall that under the Borda Count, *all* rankings are weighted and counted, not just first place. With 3 candidates, a ballot awards 3 points for 1st, 2 for 2nd, and 1 for 3rd. Thus the second-column voters could lower the Borda Count for candidate A by insincerely reversing their ranks of A and C.

Concretely, here is the count as the ballots stand:

Candidate	Sincere Borda Count	
A	$3(20) + 2(18) + 1(0) = 60 + 36$	$= 96 \quad \leftarrow \text{A wins}$
B	$3(18) + 2(20) + 1(0) = 54 + 40$	$= 94$
C	$3(0) + 2(0) + 1(20 + 18)$	$= 38$

Now suppose the 18 voters in the second column insincerely rank C above A, so their ballot reads B, C, A:

Candidate	Insincere Borda Count	
A	$3(20) + 2(0) + 1(18) = 60 + 18$	$= 78$
B	$3(18) + 2(20) + 1(0) = 54 + 40$	$= 94 \quad \leftarrow \text{now B wins}$
C	$3(0) + 2(18) + 1(20) = 36 + 20$	$= 56$

By burying A in last place, the second-column voters drop A's total from 96 to 78 while leaving B's total unchanged at 94. The winner switches from Candidate A to Candidate B, which is the outcome those voters prefer.

Problem 24 (parts a and b)

Sequential Pairwise voting is a method not commonly used for political elections, but sometimes used for shopping and games of pool. In this method, the choices are assigned an order of comparison, called an agenda. The first two choices are compared. The winner is then compared to the next choice on the agenda, and this continues until all choices have been compared against the winner of the previous comparison.

Number of voters	10	15	12
1st choice	C	A	B
2nd choice	A	B	D
3rd choice	B	D	C
4th choice	D	C	A

(a) Using the preference schedule above, apply Sequential Pairwise voting to determine the winner, using the agenda: A, B, C, D.

Solution

We work down the agenda one comparison at a time.

Step 1: Compare A and B.

Candidate	# of voters preferring
A	$10 + 15 = 25$
B	12

A is preferred, so A advances.

Step 2: Compare A and C.

Candidate	# of voters preferring
A	15
C	$10 + 12 = 22$

C is preferred, so C advances.

Step 3: Compare C and D.

Candidate	# of voters preferring
C	10
D	$15 + 12 = 27$

D is preferred.

The agenda is exhausted, so the winner is Candidate D.

(b) Show that Sequential Pairwise voting can violate the Pareto criterion.

*Pareto Criterion: If **every** voter prefers candidate A to candidate B, then B should not be the winner.*

Solution

Look at how each group of voters ranks B relative to D:

Number of voters	10	15	12
Rank of B	3rd	2nd	1st
Rank of D	4th	3rd	2nd

In every single column, B is ranked above D. So *every* voter — all 37 of them — prefers candidate B to candidate D. Yet by part (a), D is the winner.

Since every voter prefers B to D and D still wins, the Pareto criterion is violated.

Problem A

(1) Explain Arrow's Impossibility Theorem in your own words in such a way that a typical adult could understand it.

Solution

No matter what voting scheme is used, there may occur outcomes that don't feel fair.

More fully: mathematicians have written down a short list of properties that any reasonable voting method "ought" to have — for example, if every voter prefers one candidate to another, the second one shouldn't win, and dropping a hopeless candidate from the race shouldn't change who wins. Arrow's theorem says that when there are three or more candidates, no ranked voting method can satisfy all of those properties at once. Every method has to give something up. So the choice of a voting system is not a matter of finding the one perfect method — there isn't one — but of deciding which kinds of unfairness we are most willing to live with.

(2) The President of the United States is elected by the Electoral College. Now that you have learned a lot about voting schemes, what scheme would **you** choose to elect the President of the United States and why?

Solution — many answers are acceptable

Full credit requires naming a specific scheme and giving reasons that draw on the properties we studied. A sample response:

I would use Instant Runoff Voting with a national popular vote. Under the current system, a third-party candidate can act as a spoiler — as in #20 of Homework 1, where candidate C's entry flipped the winner from B to A. IRV largely removes that problem, because if my favorite candidate is eliminated my vote transfers to my next choice, so I never have to vote insincerely for the "lesser evil." It also guarantees the winner has majority support after transfers, which plurality does not. I know IRV is not perfect — it can violate monotonicity, as we saw — but I would rather accept that rare failure than keep a system where voting sincerely can hurt the candidate you most want to win.

Other well-argued answers — keeping the Electoral College, Borda Count, approval voting, a Condorcet method — earn full credit if the reasoning is specific and uses the vocabulary from this unit.

(3) How has learning about voting schemes changed your view of voting and elections?

Solution — many answers are acceptable

Full credit requires a specific, reflective response that connects to something actually studied in this unit. A sample response:

I used to think that counting votes was the easy, objective part of an election and that any disagreement was about the politics. What surprised me is that the counting method itself decides the outcome: in #6 of Homework 1, the very same 47 ballots produced Standard A under plurality, Standard B under Borda Count, Standard A under IRV, and a tie between B and D under Copeland. Nobody changed their mind — only the rule changed. That makes me pay much more attention to *how* an election is run, and it makes me more sympathetic to arguments about election reform than I was before.

Part II: Weighted Voting

Problem 2

Consider the weighted voting system $[31 : 10, 10, 8, 7, 6, 4, 1, 1]$.

(a) How many players are there?

Solution

The numbers after the colon are the players' weights, and there are 8 of them. So there are 8 players.

(b) What is the total number (weight) of votes?

Solution

Add the weights:

$$10 + 10 + 8 + 7 + 6 + 4 + 1 + 1 = 47.$$

The total weight is 47 votes.

(c) What is the quota in this system?

Solution

The quota is the number before the colon: $q = 31$. For context, this is $\frac{31}{47} \approx 0.6596$, or about 66% of the total weight.

Problem 4

Consider the weighted voting system $[q : 10, 9, 8, 8, 8, 6]$.

(a) What is the smallest value that the quota q can take?

Solution

First find the total weight of votes:

$$10 + 9 + 8 + 8 + 8 + 6 = 49.$$

The quota must be more than half the total weight; otherwise two opposing coalitions could both reach the quota at the same time. Since

$$\frac{1}{2}(49) = 24.5,$$

we need $q \geq 25$, so the smallest quota is $\boxed{q = 25}$.

(b) What is the largest value that the quota q can take?

Solution

The quota cannot exceed the total weight of 49 — otherwise no coalition, not even all the players together, could ever pass a motion. So the largest quota is $\boxed{q = 49}$.

(c) What is the value of the quota if at least two-thirds of the votes are required to pass a motion?

Solution

Two-thirds of the total weight is

$$\frac{2}{3}(49) = 32.\bar{6}.$$

Since “at least two-thirds” must be met with a whole number of votes, we round up: the quota would be $\boxed{q = 33}$.

Problem 5

Consider the weighted voting system $[13 : 13, 6, 4, 2]$.

(a) Identify the dictators, if any.

Solution

Player 1 (call this player P_1) has weight equal to the quota, $13 = 13$. So P_1 can pass any motion alone, with no help from anyone: $\boxed{P_1 \text{ is a dictator}}$.

(b) Identify players with veto power, if any.

Solution

P_1 has veto power, since a dictator automatically has veto power — the remaining players have combined weight $6 + 4 + 2 = 12 < 13$, so nothing passes without P_1 . No one else has veto power.

(c) Identify dummies, if any.

Solution

Players P_2 , P_3 , and P_4 are dummies. If there is a dictator, all other players are dummies: every winning coalition already contains P_1 , and removing any of P_2 , P_3 , P_4 leaves a coalition containing P_1 , which still wins. So none of them is ever critical.

Problem 6

Consider the weighted voting system $[11 : 9, 6, 3, 1]$.

(a) Identify the dictators, if any.

Solution

The total weight here is $9 + 6 + 3 + 1 = 19$, and the quota is 11.

There is no dictator, since all the weights are below the quota: $9 < 11$, $6 < 11$, $3 < 11$, $1 < 11$. No player can pass a motion alone.

(b) Identify players with veto power, if any.

Solution

P_1 has veto power. The players other than P_1 have combined weight

$$6 + 3 + 1 = 10 < 11,$$

so no motion can pass without P_1 .

No one else has veto power. For instance, $\{P_1, P_3, P_4\}$ has weight $9 + 3 + 1 = 13 \geq 11$ and wins without P_2 , so P_2 has no veto; and $\{P_1, P_2\}$ has weight $15 \geq 11$ and wins without either P_3 or P_4 .

(c) Identify dummies, if any.

Solution

P_4 is a dummy. There is no winning coalition for which P_4 is critical. To see this, list the winning coalitions containing P_4 and check whether removing P_4 breaks them:

Winning coalition	Weight	Weight without P_4
$\{P_1, P_2, P_4\}$	16	$15 \geq 11$ — still wins
$\{P_1, P_3, P_4\}$	13	$12 \geq 11$ — still wins
$\{P_1, P_2, P_3, P_4\}$	19	$18 \geq 11$ — still wins

P_4 is never critical, so P_4 is a dummy.

By contrast, P_3 is *not* a dummy: the coalition $\{P_1, P_3\}$ has weight $12 \geq 11$ and wins, but without P_3 the weight drops to $9 < 11$, so P_3 is critical there.